Worksheet 6-9 B Integration with U-Substitution #2
AP Calculus Name_____

**Directions:** Evaluate the following indefinite integrals. SHOW ALL WORK.

1. $\int -6x(7-3x^2)^5\,dx$

2. $\int \frac{1}{4-x}\,dx$

3. $\int -3x^2\sqrt{x^3-4}\,dx$

4. $\int x\sqrt[3]{4+x^2}\,dx$

5. $\int (x^2+1)(x^3+3x)^4\,dx$

6. $\int r^{\frac{1}{2}}\sqrt{r^{\frac{3}{2}}+5}\,dr$

7. $\int \frac{-x}{\sqrt{2-x^2}}\,dx$

8. $\int \frac{5x^2}{(5+2x^3)^3}\,dx$

9. $\int \frac{2x^2}{x^3-1}\,dx$

10. $\int \frac{3y+2}{\sqrt[4]{3y^2+4y}}\,dy$

11. $\int \left(3-\frac{1}{t^2}\right)^4\left(\frac{5}{t^3}\right)dt$

12. $\int \frac{e^x}{4-e^x}\,dx$

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13. $\int e^{3-2x}\, dx$

14. $\int -e^{2x+6}\, dx$

15. $\int \cos^3\theta \sin\theta\ d\theta$

16. $\int xe^{x^2+1}\, dx$

17. $\int \sin(3\theta - 1)\, d\theta$

18. $\int \frac{2}{3-4x}\, dx$

19. $\int \tan x \cos x\ dx$

20. $\int \sqrt{\cot x}\csc^2 x\ dx$

21. $\int \frac{(\ln x)^5}{x}\, dx$

22. $\int \left(\cos y\, e^{\sin y} + y^2\right) dy$

23. $\int \csc(2t-1)\cot(2t-1)\, dt$

24. $\int (\sin(2x-1) + 3\cos(2x))\, dx$

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